INVINITY ENERGY SYSTEMS



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SPC100644

Endurium BMS Modbus Interface Specification

Invinity Energy Systems

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1 Introduction

This document details the interface provided by the Battery Management System (BMS) for controlling and monitoring battery operations. The BMS offers these functionalities via MODBUS TCP client and server interfaces, enabling external control devices to manage and oversee the performance and configuration of the system in a reliable and standardized manner. The system also allows multiple listeners to observe its operations.

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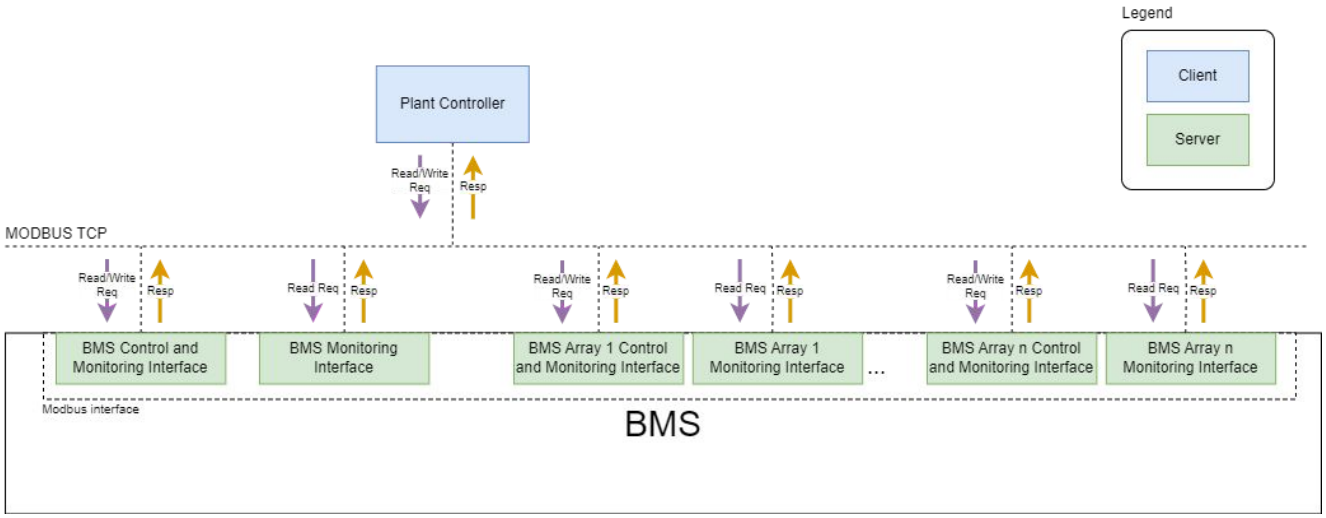
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2 Overview

The BMS functions within a client/server architecture, specifically designed to accommodate two types of BMS systems: DC-Configured BMS and AC-Configured BMS. It offers a straightforward yet robust system interface that simplifies the operation and monitoring of large battery systems.

Illustrated below is the client/server architecture, for a DC-configured BMS (where PCS devices are controlled by the Plant Controller). A Plant Controller primarily interacts with the BMS Control and Monitoring Interface (see page 9) to operate the battery system issuing Operating Commands and Power Requests. It also interacts with a secondary BMS Array Control and Monitoring Interface (see page 9) that provides per-Array operational status and information and limited control functionality.

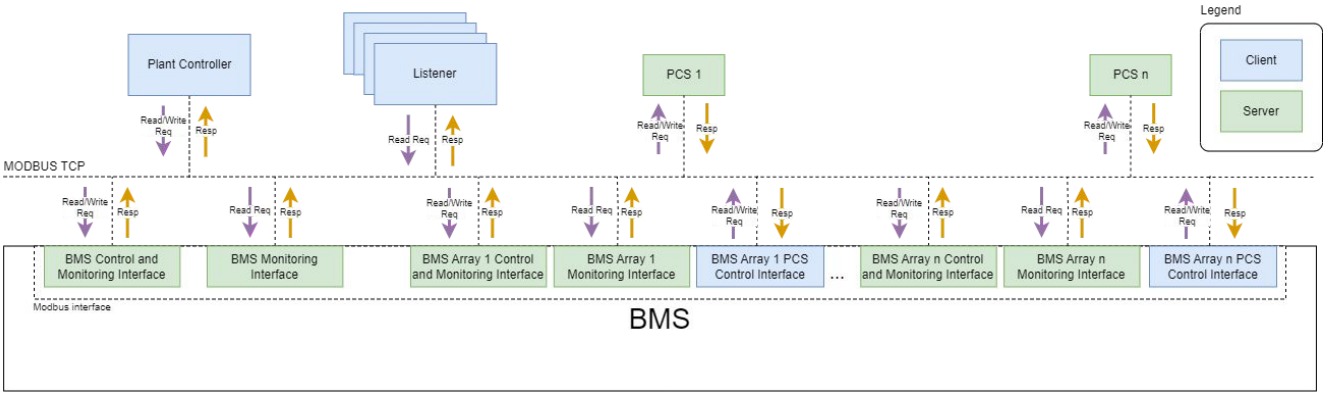
At the same time multiple Listeners are allowed to connect to the BMS Monitoring Interface (see page 9) and BMS Array Monitoring Interface (see page 9) for monitoring purposes only (no operating control is available).



For an AC-Configured BMS, the client/server architecture described above expands the interface with a Per-Array Modbus Client interface to control the PCS device associated with each Array:

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2.1 BMS Control and Monitoring Interface

The BMS enables centralized system control through a server supporting single MODBUS TCP connection. Utilizing this port enables both write and read operations to control and configure the operation of the battery system and monitor its performance.

2.2 BMS Monitoring Interface

A secondary server supports a maximum of 5 connections (5 clients) for monitoring the BMS operation on a distinct Modbus server. These connections are restricted to read-only operations for fetching various operational parameters, with no write permissions granted.

2.3 BMS Array Control and Monitoring Interface

The BMS provides a set of Modbus servers dedicated to supplying the controlling device with Array-specific operating parameters to support the operation of the Arrays. Each Array within the system is assigned an individual Modbus server supporting a single MODBUS TCP connection. While these ports enable both read and write operations, the control and configuration of the Arrays are limited, offering a lower level of controllability compared to the **BMS Control and Monitoring Interface**. This allows for basic monitoring and restricted adjustments to support Array operation.

2.4 BMS Array Monitoring Interface

Another set of secondary servers each supporting a maximum of 5 connections (5 clients) allows monitoring the BMS Arrays operation on distinct Modbus servers. Similar to the BMS Monitoring interface, these connections are restricted to read-only operations for fetching various operational parameters, with no write permissions granted.

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2.5 BMS Array PCS Control Interface:

For AC-configured systems, a per-Array Modbus client is responsible for interfacing with the Power Conversion System (PCS). The interface facilitates both control and monitoring operations.

The interface between the Battery Management System (BMS) and the Power Conversion System (PCS) enables the BMS to send control signals for adjusting power output, changing operational modes, and regulating voltage and current levels. Additionally, it retrieves essential operational data from the PCS, including real-time power metrics, energy flow status, fault codes, and performance metrics. This data is crucial for the BMS to ensure the optimal operation of the battery system.

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3 Ports Configuration

- BMS Control and Monitoring Interface:
  - Port Number: 5020 (default, configurable)
  - Connection Limit: This port accepts a maximum of 1 active connection at any given time.
  - Purpose: The control port is designated for performing writes for control and configuration of the system, as well as performing read operations to retrieve operating data parameters.
- BMS Monitoring Interface:
  - Port Number: 5030 (default, configurable)
  - Connection Limit: This port permits up to 5 simultaneous connections.
  - Purpose: The read-only port is exclusively for reading data from the BMS. Clients can request information from these ports; however, they do not have any write capabilities
- BMS Array Control and Monitoring Interface
  - Base Port Number: 1400 (default, configurable)
    - The port number for each Array is determined by adding an offset to the base port number. The offset corresponds to the Array's instance number.
  - Connection Limit: This port accepts a maximum of 1 active connection per port at any given time.
  - Purpose: The per-Array control ports are designated for performing, for each BMS Array, writes for control and configuration, as well as performing read operations to retrieve operating data parameters.
- BMS Array Monitoring Interface:
  - Base Port Number: 1500 (default, configurable)
    - The port number for each Array is determined by adding an offset to the base port number. The offset corresponds to the Array's instance number.
  - Connection Limit: Each of the BMS Array Status ports supports a maximum of 5 simultaneous connections.
  - Purpose: All BMS Array Monitoring ports are designed for reading data from the BMS Arrays. Clients can request information from these ports; however, they do not have any write capabilities. Each port corresponds to a specific array.
- BMS Array PCS Control Interface:
  - Port Number(s): Configured based on the PCS Modbus server (configurable).
  - Connection Limit: This port establishes a single connection between the BMS and the PCS.
  - Purpose: It allows the BMS to perform write operations to control the PCS and perform read operations to retrieve operational data from the PCS.
- Connection Rules:
  - Connection Process: Clients are expected to initiate a connection to the appropriate port based on their intended operations:
    - For control operations, clients should connect to the control port.
    - For data retrieval, clients should connect to either the BMS Monitoring port and/or to one or more of the BMS Array Monitoring ports.
    - Once the max number of connection is reached, further attempts to connect from a client will be refused. Clients attempting to connect beyond this limit will receive an appropriate error message, indicating that the maximum connection threshold has been exceeded.
  - Timeout and Disconnection Handling: The servers will automatically close connections that remain inactive for a configured duration. In such cases, clients are required to reconnect. See **Connection Monitoring** for details.

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4 Communication Details

4.1 Supported Registers

The following register types are supported:

- Input Registers
- Holding Registers

4.2 Function Codes Supported

- Read Holding Registers (0x03)
- Read Input Registers (0x04)
- Write Single Register (0x06)
- Write Multiple Registers (0x10)
- Read/Write Multiple Registers (0x17)

4.3 Exception Handling

Standard Modbus exception codes serve as indicators of errors that occur during interactions between a Client and the server.

Error Code	Description	Example
Illegal Function (Code 01)	The function code received in the query is not supported by the server.	A client attempts to write to a register on a read-only Modbus interface (e.g., trying to write to a read-only data port).
Illegal Data Address (Code 02)	The data address specified in the query is not valid in the server.	The client requests data from a register that is outside the defined range of the BMS's Modbus map.
Illegal Data Value (Code 03)	The value in the query data field is not acceptable by the server.	A client tries to write a setpoint that is outside the allowable range (e.g., attempting to set the battery voltage beyond safe limits) or the number if read/write registers in the requests exceeds the maximum allowed.

4.4 Data Types and Format

Name	Description	Range
int16	16-bit signed integer (1 word)	-32768...+32767
uint16	16-bit unsigned integer (1 word)	0...65535
int32	32-bit signed integer (2 words)	-2 147 483 648...+2 147 483 647

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Name	Description	Range
uint32	32-bit unsigned integer (2 words)	0...4 294 967 295
float32	32-bit value (2 words)	-3.4028E+38... +3.4028E+38

16-bit variables are transmitted with Big Endian format:

- The most significant byte is transmitted first.
- The least significant byte is transmitted second.

32-bit variables are transmitted with Big Endian format:

- The most significant word is transmitted first, then the least significant.

4.5 Connection Monitoring

4.5.1 Control Connection

The BMS Status and Control Port provides the clients with a heartbeat mechanism to maintain the connection alive beyond read and write request.

Heartbeat Message

A heartbeat mechanism is employed to maintain the connection active. This requires the client to regularly monitor the BMS heartbeat signal, suggested at a 1-second interval and ‘echoing’ back via a heartbeat write request. The received heartbeat must increment from the previous one. Any successive heartbeat writes with the same value will be disregarded, and the timeout mechanism will keep counting up. The heartbeat parameter is updated at a 1s rate.

4.5.2 Monitoring Connection

For read-only interfaces like the BMS Status and BMS Array Ports, the clients keep connections active by regularly sending read requests prior to the set timeout timer elapsing.

4.5.3 Timeout Response

If the client does not maintain the connection as described, the BMS will close the connection after the configured inactivity timeout and the client must reconnect.

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5 Performance and Limitations

5.1 Maximum Registers Per Request

Up to 25 registers can be read per request, and up to 25 registers can be written per request.

5.2 Load Management

Each Modbus server in the BMS continuously monitors the total register access activity across all clients. Instead of monitoring individual client requests, the system evaluates the overall load on each server to ensure that access limits are respected. Each Modbus server has a predefined threshold of 50 register accesses allowed per 1 second (on average). This threshold is established to prevent excessive load and maintain optimal system performance.

5.2.1 Request Handling When Threshold is Surpassed

If the total number of register access requests to a Modbus server exceeds the defined threshold within the monitoring period, any additional requests are automatically dropped by the server. Dropped requests will not be queued or retried by the server, and clients will need to manage retry attempts as necessary.

5.2.2 Excessive Polling Flag

A flag is provided to alert the client when polling is excessive to allow the client to adjust its behavior accordingly. This flag is provided through a Modbus register and supports the following values:

- 0 = No excessive polling
- 1 = Excessive polling

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6 Operational Instructions

6.1 BMS Control Modes

The BMS supports two distinct control modes across all four control layers: System, Array, String, and Module controllers.

- At the System Control level, the BMS functions as follows:
  - **Normal Control Mode**: the Modbus interface facilitates communication and control, allowing a Plant Controller to manage the BMS through its system parameters and monitor performance directly through this protocol. While in this mode, all Modbus commands are fully operational, granting the operator the ability to execute commands, read data, and make adjustments as needed.
  - **User-Controlled Mode:** while in this Mode the Modbus interface will ignore all write requests (accept but not store the value received) for controlling the system, and all operational control is transferred to an operator via a dedicated web-based service interface. This mechanism ensures that the system is managed securely and effectively through the web interface, preventing any potential conflicts or unauthorized modifications that may arise from simultaneous access through the Modbus protocol.

  The control mode is managed through a dedicated web-based service interface, enabling users to take control of the system for tasks such as maintenance or troubleshooting. Once the designated tasks are completed, users can return the system to its normal state.

  The active control mode is provided through Modbus register **Control Mode [Reg 31036]**.

- For the Array, String, and Module control layers, transitioning from normal mode to user-controlled mode signifies that the component becomes unavailable to the upper-layer controller for normal operation. This change is observable at the system level, resulting in a decrease in operational capabilities until the component is reverted to normal mode, allowing the parent controller to regain control.

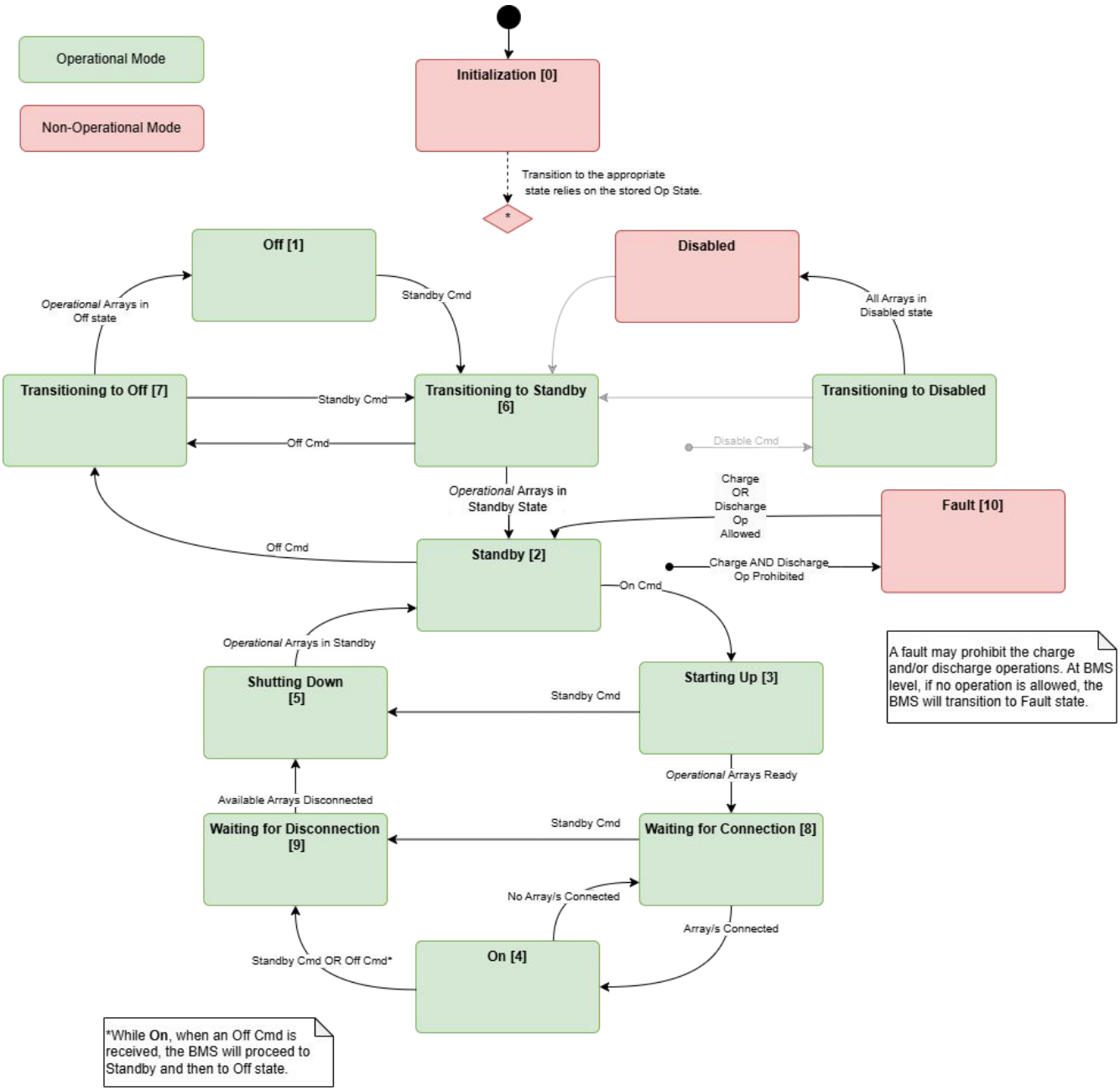
  If an Operator requests control over an Array, that control is transferred to the user, and the System Controller no longer considers it for operation. Once the Operator releases control of the Array, the System Controller automatically requests and regains control, resuming normal operation of that component.

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6.2 BMS Operation

The BMS is operated through the **Operating Command [Reg 41000]**. The following diagram shows the Top-level state machine governing the operation of the system:



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6.2.1 Battery in Standby

In Standby all auxiliary functions designed to keep operating conditions within normal limits (e.g., freeze prevention) are permitted to operate; however, the Battery itself is not ready for active use. It stays in Standby, awaiting the ‘On’ command to initiate the startup process.

6.2.2 Turning the Battery On

The Controller is required to issue an ‘On' command to start up the system and the Arrays. The BMS is considered ‘On’ when at least one Array has successfully connected one of its String to the DC Bus and, for AC-configured systems, the PCS is connected/running.

When ‘On’ the BMS will report back through the **Operating State [Reg 31000]** the current operation depending on the operating conditions: ‘Idle’ (when at rest), ‘Charging’, or 'Discharging’.

Important: It is highly recommended that the Plant Controller enforces a delay of at least 5 second after the BMS reports the battery as On following an On command, before issuing any non-zero charge/discharge power setpoint.

- This delay is critical to allow String connection to complete and stabilize around the DC Bus voltage. String contactor qualification takes approximately 1 second; allowing this time enables the system to connect all available Strings so they can actively participate in the charge or discharge operation.
- If non-zero power is commanded immediately after the system reports On, the resulting voltage difference between the Strings and the DC Bus may prevent additional Strings from connecting. Those Strings may then be left out of operation until conditions allow them to connect.
- At the Array level, the Modbus interface exposes both the number of connected Strings and the number of available Strings. Integrators can use these values to verify that the String connection process is progressing as expected. If the Strings within an Array are relatively balanced, all available Strings should connect during readiness/startup, subject to operating conditions and limits.

6.2.3 Turning the Battery Off

The battery can be turned Off from Standby. When in this state, all auxiliary functions are disabled, effectively halting internal operations. This design aims to minimize energy consumption to the lowest possible level.

6.2.4 Shutting Down the Battery

When the Battery System is in the 'On' state, the Controller has the capability to issue a Standby command. This command prompts the system to disconnect and transition all operating Arrays to Standby. Before this disconnection occurs, the **Charge Power Limit [Reg 31019]** and **Discharge Power Limit [Reg 31021]** are both set to 0, to halt all power flow.

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6.2.5 Battery System Disabled

The Battery System may be disabled if it is not in a condition to operate, such as during major maintenance activities. When disabled, its behavior mirrors that of the Off state: all auxiliary functions are turned off, effectively halting internal operations. It is important to note that disabling and re-enabling the Battery System is not possible through the Modbus interface; this action can only be performed via the Service Interface.

6.2.6 Power Readiness Request

The BMS optimizes the operation of the Arrays and their corresponding Strings to effectively manage the power levels specified by the **Charge Readiness Power Command [Reg 41003]** and the **Discharge Readiness Power Command [Reg 41005].** In the simplest case, the operator can specify the total power capacity of the system with this command. If more fine-grained, efficient usage is required, the operator can specify lower values. Charge and Discharge commands are provided so that the BMS can order the connection of strings in the most effective manner to meet the requirements.

When a power level is requested for charging and/or discharging, the BMS disaggregates the total power requested, distributing it across the available Arrays in the system. This distribution process ensures that the system operates within safe limits and that the workload is balanced across the Arrays.

The readiness involves ramping up the necessary Strings in the battery system to ensure that when the operation begins, it can efficiently handle the required power. Operating conditions may prevent all necessary Strings required to match the **Readiness Power** level to start immediately. The BMS continuously monitors the operating conditions and will command the available Strings into operation when possible.

The **DC Power Target [Reg 41007]** enables the Plant Controller to communicate the anticipated operational power level the battery system will be subjected to. This forward-looking information is critical during the battery's startup, as it allows the Battery Management System (BMS) to optimize its preparation for the upcoming charge or discharge operation.

Operating Limits

The Operating Limits outlined in this section are critical parameters that must be adhered to at all times by an external control device interfacing with the BMS. These limits ensure the safe and efficient operation of the battery system, preventing potential damage and optimizing performance. By respecting these boundaries, operators can maintain the integrity of the system while facilitating effective control and monitoring of the battery's operational capabilities.

1 BMS Operating Limits

As the Arrays become online and Strings are connected the battery will update the **Charge Power Limit [Reg 31019]** and **Discharge Power Limit [Reg 31021]** as well as the **Charge Energy [Reg 31017]** and **Discharge Energy [Reg 31015]**.

2 BMS Array Operating Limits

Each Array, similar to how the System reports its operating limits, communicates its operational boundaries via its Modbus interfaces. This is accomplished through the **Charge Power Limit [Reg 31017]** and **Discharge Power Limit [Reg 31019]** registers, in addition to the **Charge Energy [Reg 31015]** and **Discharge Energy [Reg 31013]** registers.

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6.2.7 Operating Power Limits Permitted

The Controller can command the Battery System utilizing the **Operating Charge Power Limit Permitted [Reg 41014]** and **Operating Discharge Power Limit Permitted [Reg 41016]** registers to restrict its charge and/or discharge capabilities, thereby ensuring that the system does not exceed a specified power level. This functionality is essential for preventing operations beyond certain limits during the commissioning phase, or maintenance or servicing of the system.

This command is only relevant for AC batteries.

6.2.8 Alert Management

The Battery System continuously monitors its operations and generates Alerts to signal any abnormal conditions or deviations from nominal ranges. There are 3 different types of Alerts which are reported asynchronously as they happen:

- **Warning**: indicates there is an abnormal condition in the Battery System, but it can continue operating in some capacity. Performance levels may be reduced or control strategies adapted to accommodate the abnormal condition. The warning may clear on its own, or it may need some Operator action for the condition to be removed. Once the condition that caused the warning is no longer present, the warning will be cleared and the system will resume operating as normal.
  - Example:

    Fan failure: a fan that cannot run prevents the unit from cooling down but does not prevent the operation of the module (until the temperature rises beyond acceptable levels at least). This is considered a warning to the Operator to indicate such a situation but does not affect the operation.
- **Fault**: indicates an event or condition that prevents the Controllers from continuing operation and the system will attempt to stop the operation in a safe manner. If the event or condition that caused the fault goes away, the system is allowed to resume operation.
  - Example:

    Electrolyte Over-temperature: in the event of an over-temperature in the electrolyte the operation needs to be limited to prevent further temperature escalation and possible damage. This alert is classified as a fault that limits the operation by not allowing charging. Once the temperature decreases, the operation can be resumed.
- **Latching Fault**: indicates the Battery System cannot continue operating safely and proper shutdown actions are required. The operation does not resume automatically. To resume operation, the conditions that caused the fault must go away and the Operator must send a **Clear Alerts [Reg 41001]** command.
  - Example:

    Pump Motor Failure: This issue is classified as a critical failure within a module. It is categorized as a latching fault, meaning that no operations are permitted until the module has been properly serviced.

The following table summarizes the response of the system to the different type of alerts:

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Type	Prohibits
Warning	All-permitted: All subsystems are allowed to operate, perhaps with reduced performance depending on the condition.
Fault	Prohibits defined are based on the nature of the condition detected: • Charge prohibited - Charge Power Limits set to 0. • Discharge prohibited - Discharge Power Limits set to 0.
Latching Fault	All-prohibited: total shutdown. No subsystem is allowed to operate.

Alert Reporting

The alert registers available in the Modbus interface serve to monitor and report the operational status of both the battery system and its individual components. Their primary purpose is to provide notifications regarding any anomalies or faults that may arise during operation.

The alert registers are specifically designed to retain their values until they are explicitly accessed by the user. This mechanism guarantees that alerts are effectively communicated once they become active, thereby preventing the loss of critical information—particularly for alerts that may activate and deactivate quickly.

For each Array, at String level a dedicated bit-field register enables users to identify which strings within the array has experienced a change in an alert state. This functionality helps prevent unnecessary polling of all strings' registers, allowing users to focus only on those that require attention.

BMS Alert Register

The **BMS Alert Register [Reg 31039]** provides a summary of the alert states at the System level. This register allows for quick assessment of the overall health and operational status of the System.

BMS Aggregated Array-Level Alerts

The **BMS Aggregated Array-Level Alerts [Reg 31041]** provides a summary of the alert states at the BMS level that are the result of Array alert aggregation.

Array-Specific Alert Register

The **Array-Specific Alert Bit-Field [Reg 31036]** provides a summary of the alert states at the Array level. This register allows for quick assessment of the overall health and operational status of the Array.

Array Aggregated String-Level Alerts

The **Array Aggregated String-Level Alerts Register [31038]** provides a summary of the alert states at the Array level that are the result of String alert aggregation.

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String Alert Change Register Summary

String Alert Change Summary [Reg 31040] serves as a mechanism to efficiently monitor the alert status of each String within the Array. By indicating which Strings have experienced changes in their alert states, it helps prevent the loss of critical information and ensures timely responses to any issues that may arise.

String Alert Registers

Each String has its own dedicated alert register, allowing users to assess the operational status and health of every String independently.

6.3 Grid Modes and AC Power Support

For AC-configured battery systems, the BMS provides the Modbus TCP interface that a plant controller uses to command grid services. The BMS then acts as master to each PCS (one per Array), translating System/Array-level commands into PCS controls while enforcing operating limits and availability. This section explains how integrators should use the interface to select scope, choose grid modes, pick active/reactive power control modes, set setpoints, and verify operation.

- **Control Status [Reg 31043]:**
  - Bitfield summarizing high-level states. For integration and troubleshooting prefer reading detailed status/ measurements; use this only as a quick summary per Modbus map note.
- **Grid Control Scope Select [Reg 41018]:**
  - Select System-level vs per-Array override. System scope applies a single set of commands to all Arrays; Array scope allows per-Array modes. Choose System for uniform dispatch; choose Array when Arrays must differ.
- **Grid Mode Command [Reg 41019]:**
  - Set high-level interaction: e.g., 1 = Grid-Feeding (export/import per setpoints and modes).
- **Active Power Control Mode Command [Reg 41020]:**
  - 0: Invalid; 1 = Fixed P (dispatch to P setpoint); 2 = P–f droop (frequency response per configured droop curve).
- **Reactive Power Control Mode Command [Reg 41021]:**
  - 0: Invalid; 1 = Fixed Q (dispatch to Q setpoint); 2 = Power Factor (dispatch to PF target).

6.3.1 Scope selection behavior

System vs Array scope: When **Grid Control Scope Select [Reg 41018]** selects System scope, writes to **Grid Mode Command [Reg 41019]**, **Active Power Control Mode Command [Reg 41020]**, and **Reactive Power Control Mode Command [Reg 41021]** apply uniformly to all Arrays. When Array scope is selected, those same commands are taken from each Array's interface, allowing Arrays to differ. Change of scope does not alter existing limits; the BMS continues to enforce per-Array availability and power limits.

6.3.2 Control modes for grid services

Grid-Feeding **[Reg 41019=1]**: Enables export/import based on active/reactive modes. Typical uses: energy dispatch, ancillary services.

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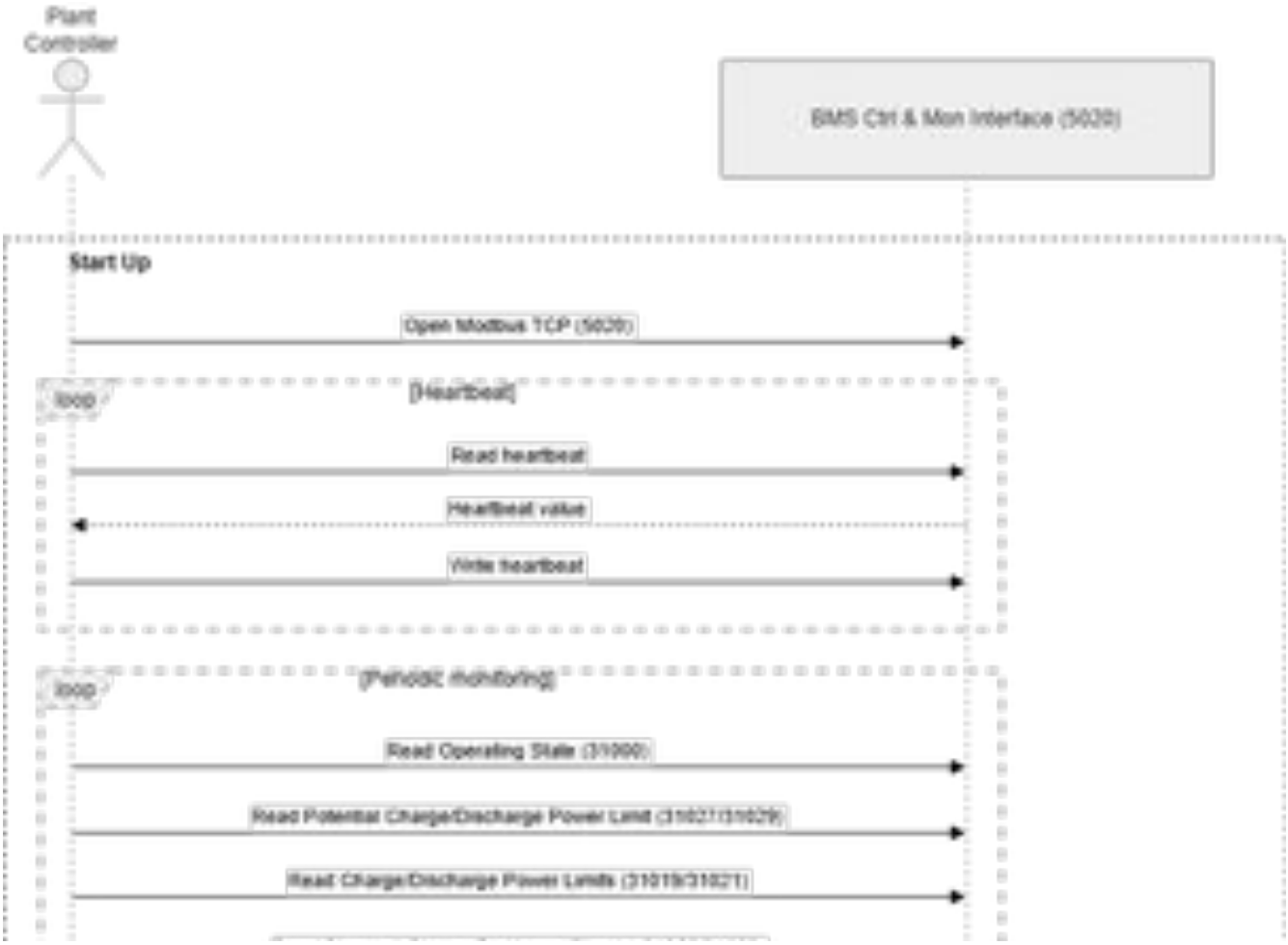
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Active Power Control Mode Command [Reg 41020]: 1 = Fixed P for energy dispatch; 2 = P–f droop to provide primary frequency response. Droop tuning parameters are configured elsewhere; this register only selects the mode.

Reactive Power Control Mode Command [Reg 41021]: 1 = Fixed Q for voltage support/VAR dispatch; 2 = Power Factor for compliance with PF targets or tariff requirements. This register selects behavior only; associated setpoints/limits and sign conventions (leading/lagging, inductive/capacitive) are defined elsewhere in the Modbus map.

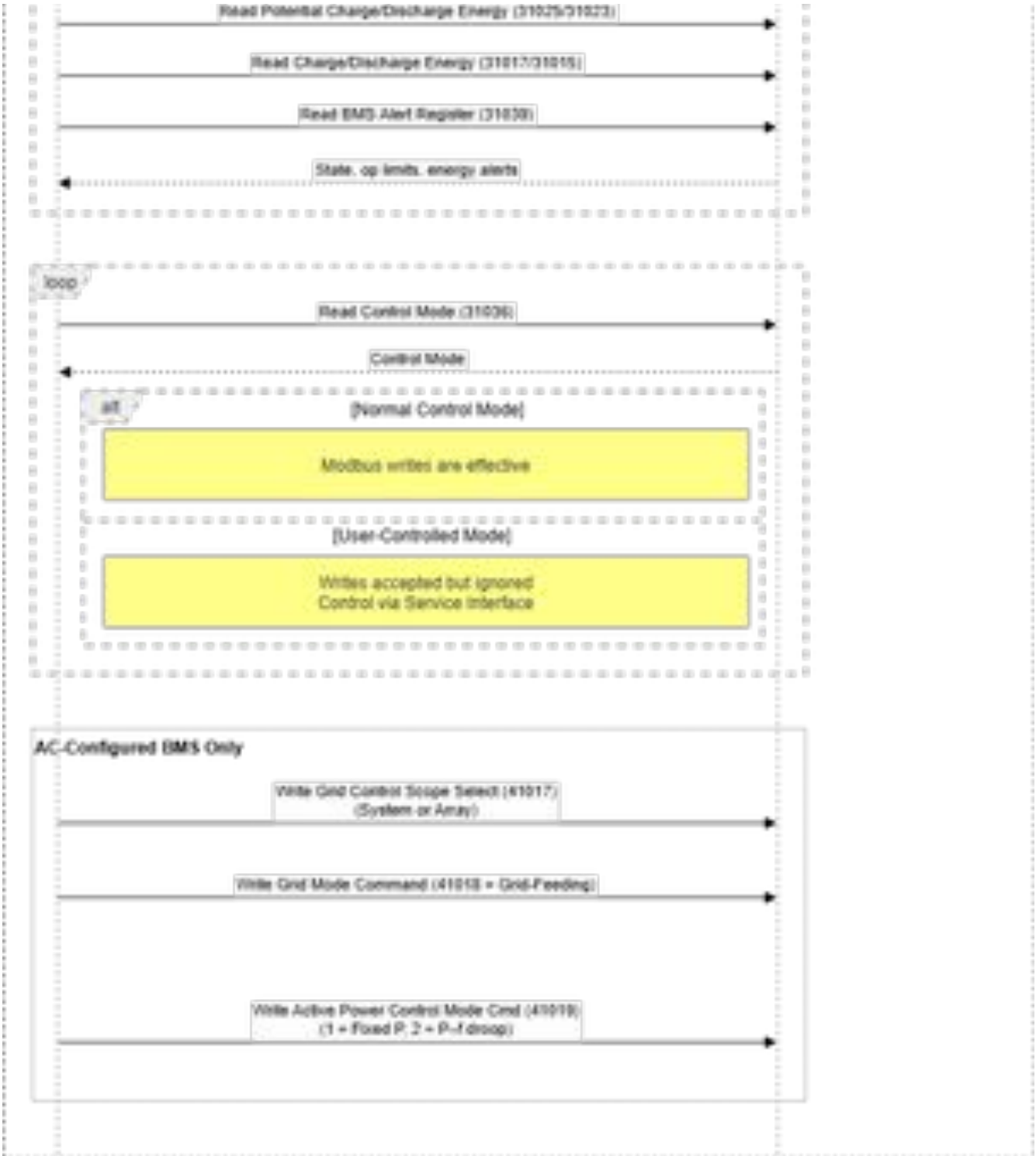
6.4 BMS Operation Sequence Diagram

The sequence diagram below summarizes typical Modbus interactions between a Plant Controller and the BMS during normal operation. It highlights the recommended command and status flow to bring the battery from Standby to On, prepare power readiness, issue charge/discharge targets, and return the system to Standby/Off. Use it as a quick reference for message ordering and which interfaces/register groups are involved at each step.



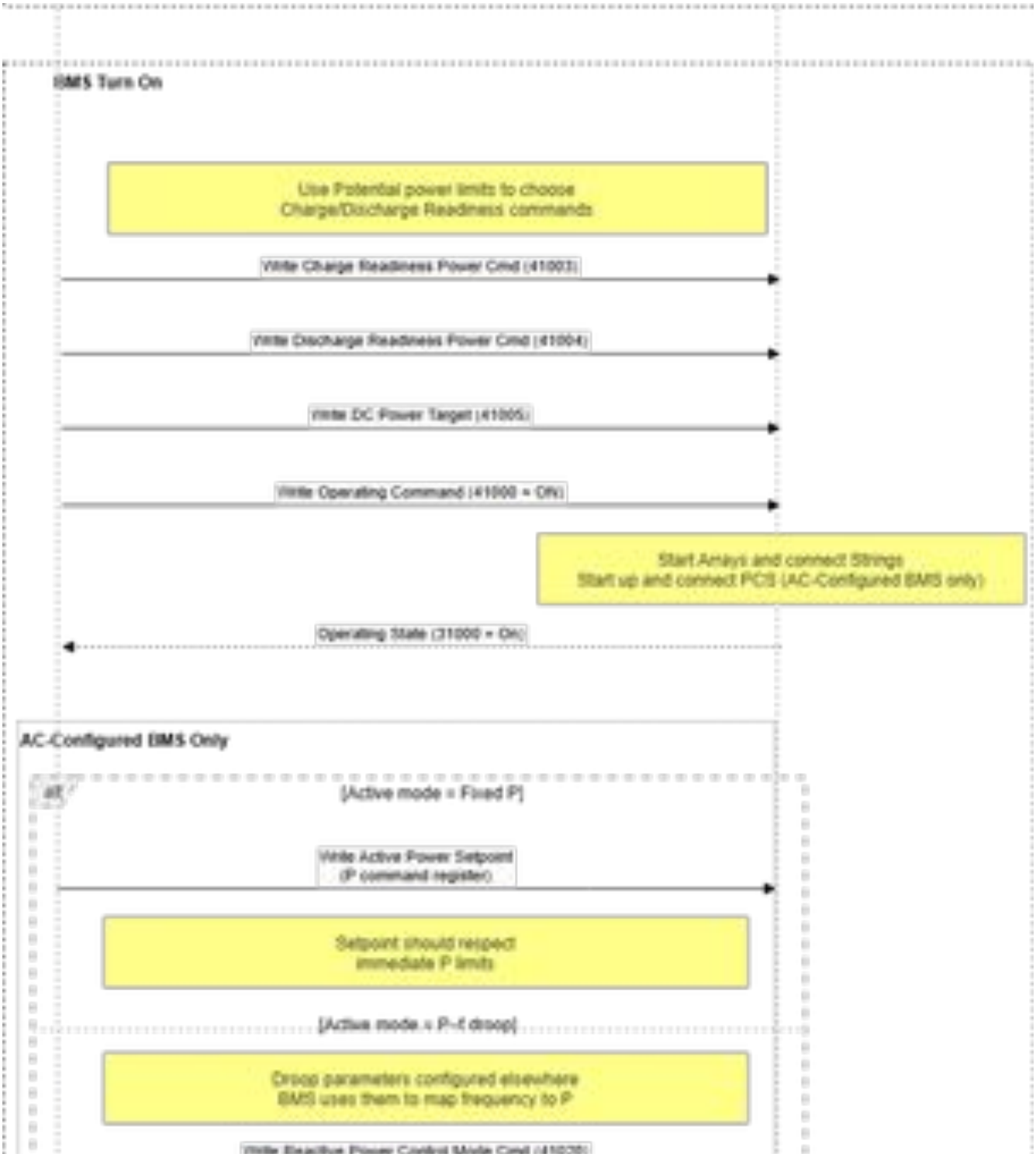
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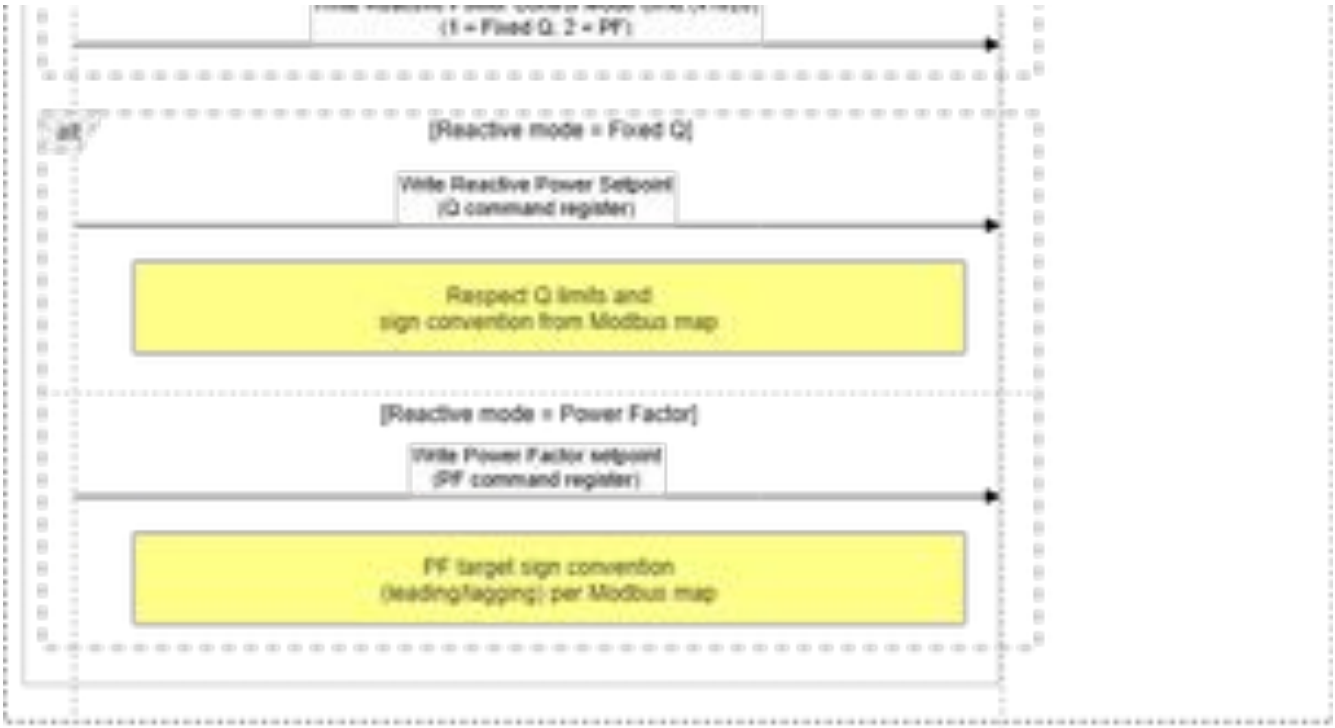
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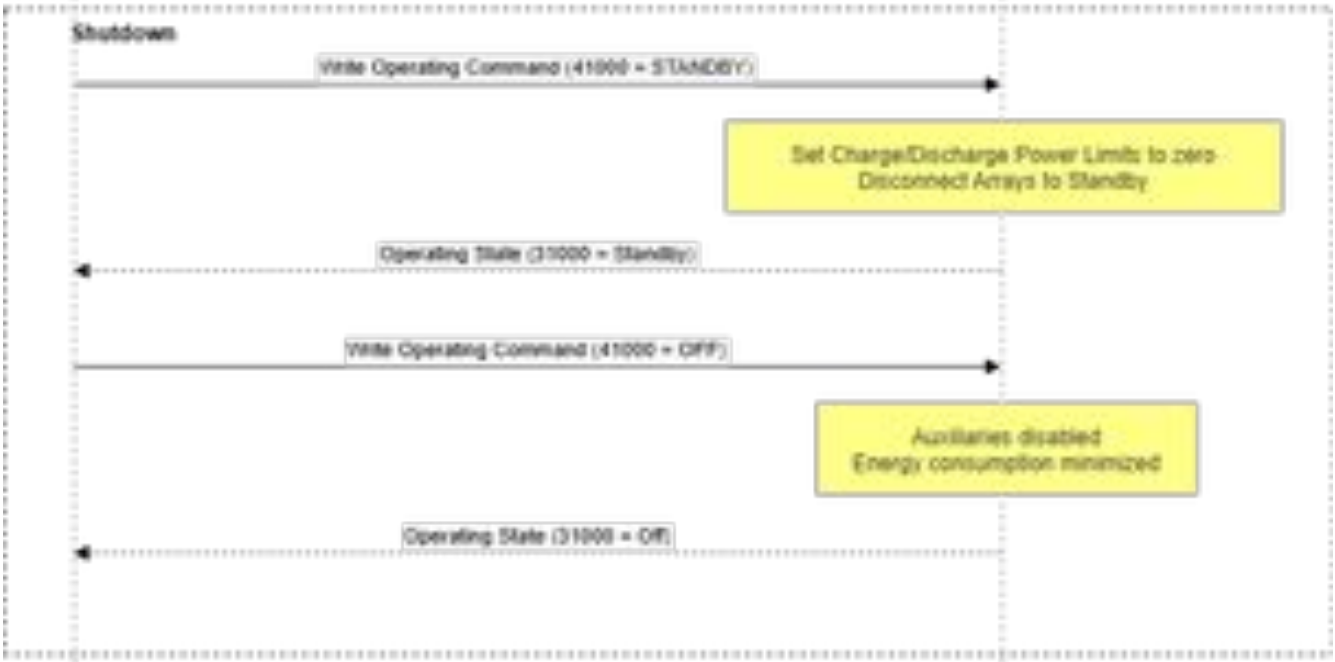
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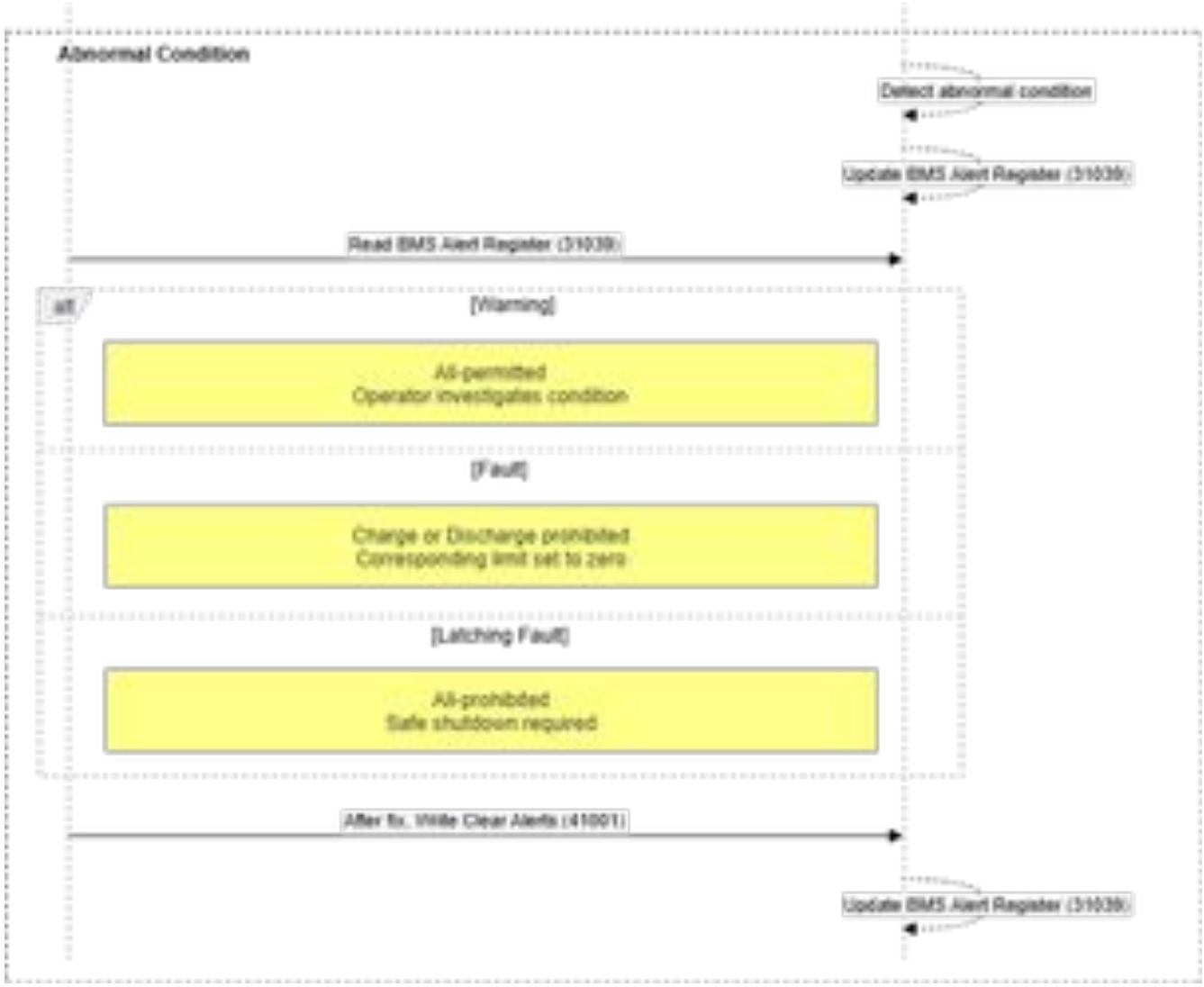
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7 Modbus map

7.1 Addressing space

7.1.1 BMS Interface

- **Input Registers (3xxxx)**: Read-only registers (16-bit):
  - Status Registers (31xxx)
  - Extended Status Registers (32xxx)
- **Holding Registers (4xxxx)**: Write/Read registers (16-bit).
  - Control Registers (41xxx)

7.1.2 BMS Array Interface

- **Input Registers (3xxxx)**: Read-only registers (16-bit)
  - Status Registers (31xxx)
  - Extended Status Registers (32xxx)
  - Service Registers (33xxx)
- **Holding Registers (4xxxx)**: Write/Read registers (16-bit).
  - Control Registers (41xxx)

7.2 Register Map

The address are defined utilizing zero-based addressing. For example, holding register 40001 would be transmitted as address 1.

Most registers apply to both DC and AC configured systems. Any exceptions are clearly identified.

7.2.1 Scaling

Values reported by the BMS are scaled with the factor specified in the map. The client is required to apply the inverse scale value to return the value to the unscaled state.

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7.2.2 BMS Interface

BMS Status Registers

Reg (dec)	Addr (dec)	Name	Description	Units	Type	Scale	Notes
31000	1000	Operating State	Operating state of the Battery System.	Enum	int16	n/a	0: Init 1: Off 2: Standby 3: Starting up 4: On 5: Shutting Down 6: Transitioning to Standby 7: Turning Off 8: Waiting for Connection 9: Waiting for Disconnection 10: Fault
31001	1001	Operating Status	Operating status of the Battery System.	Enum	int16	n/a	0: Invalid 1: Normal 2: Warning 3: Fault 4: Critical Fault
31002	1002	Operating Mode	Operating mode of the Battery System.	Enum	int16	n/a	0: Invalid 1: Non-Operational 2: Operational 3: Service 4: Initial Charge
31003	1003	Heartbeat	BMS Heartbeat.	Num	uin16	x1	Incremented every 1s and wraps around on overflow.

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Reg (dec)	Addr (dec)	Name	Description	Units	Type	Scale	Notes
31004	1004	Active State of Charge	Active state of charge of the Battery System.	%	uint16	x10	0 - 100% Battery system state of charge based on the Arrays actively engaged in operation (excludes available but inactive Arrays).
31005	1005	Array Maximum State of Charge	Highest State of Charge among the Battery System Arrays.	%	uint16	x10	0 - 100%
31006	1006	Array Minimum State of Charge	Lowest State of Charge among the Battery System Arrays.	%	uint16	x10	0 - 100%
31007	1007	DC Power*	Total DC power reported by the Arrays.	kW	int32	x1	$P > 0 \rightarrow Discharging$ $P < 0 \rightarrow Charging$
31009	1009	Active Power**	Total Battery System AC active power reported by the PCS devices.	kW	int32	x1	$P > 0 \rightarrow Exporting/Discharging$ $P < 0 \rightarrow Importing/Charging$
31011	1011	Reactive Power**	Total Battery System AC reactive power reported by the PCS devices.	kVAR	int32	x1	$Q > 0 \rightarrow Inductive\ Power$ $Q < 0 \rightarrow Capacitive\ Power$
31013	1013	Power Factor**	System-level ratio of active power to apparent power.	Num	int16	x100	$PF < 0 \rightarrow Leading$ $PF > 0 \rightarrow Lagging$
31015	1015	Discharge Energy	Energy level of the Battery System available for discharge.	kWh	uint32	x1	

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Reg (dec)	Addr (dec)	Name	Description	Units	Type	Scale	Notes
31017	1017	Charge Energy	Energy level required to reach Top-of-Charge in the Battery System. (From the current SOC level).	kWh	uint32	x1	
31019	1019	Charge Power Limit	Charge power immediately available from the Battery System.	kW	int32	x1	$P_{Charge} < 0$
31021	1021	Discharge Power Limit	Discharge power immediately available from the Battery System.	kW	int32	x1	$P_{Discharge} > 0$
31023	1023	Potential Discharge Energy	Energy level of the Battery System that can be available within a short period of time.	kWh	uint32	x1	
31025	1025	Potential Charge Energy	Energy level required to reach Top-of-Charge that can be available from the Battery system within a short period of time.	kWh	uint32	x1	
31027	1027	Potential Charge Power Limit	Charge power that can be available from the Battery system within a short period of time.	kW	int32	x1	$P_{Charge} < 0$
31029	1029	Potential Discharge Power Limit	Discharge power that can be available from the Battery system within a short period of time.	kW	int32	x1	$P_{Discharge} > 0$
31031	1031	Average Internal Temperature	Average temperature in the Battery System.	°C	int16	x10	

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Reg (dec)	Addr (dec)	Name	Description	Units	Type	Scale	Notes
31032	1032	Minimum Internal Temperature	Minimum temperature in the Battery System.	°C	int16	x10	
31033	1033	Maximum Internal Temperature	Maximum temperature in the Battery System.	°C	int16	x10	
31034	1034	Arrays Active Count	Number of Arrays actively supporting the operation.	Count	uint16	x1	
31035	1035	Arrays Available Count	Number of Arrays available to support the operation including those actively doing so.	Count	uint16	x1	
31036	1036	Control Mode	The control mode indicates whether the system is being controlled from the Plant Controller interface (Modus interface) or in Service under control of Service Personnel from the Service interface.	Enum	uint16	n/a	0: Normal 1: User-Controlled
31037	1037	System Id	Number identifying the System.	Num	uin16	x1	
31038	1038	BMS Type	Configured BMS type: AC or DC System.	Enum	uint16	n/a	0: DC-Configured BMS 1: AC-Configured BMS
31039	1039	BMS Alerts	Provides a summary of BMS alert states.	Bitfield	uint32	n/a	1: active alert 0: inactive alert
31041	1041	BMS Aggregated Array Alerts	Provides a summary of alert states at the System level, resulting from the aggregation of Array alerts.	Bitfield	uint32	n/a	1: active alert 0: inactive alert

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Reg (dec)	Addr (dec)	Name	Description	Units	Type	Scale	Notes
31043	1043	Control Status**†	Bitfield of battery control status flags	n/a	uint16	n/a	Bit 0 = Available (0: False, 1: True) Bit 1 = Operating (0: False, 1: True) Bit 2 = Warning (0: False, 1: True) Bit 3 = Failure (0: False, 1: True) Bit 4 = Remote Control Available (0: False, 1: True)
31044	1044	Available State of Charge	Available state of charge of the Battery System.	%	uint16	x10	0 - 100% Battery system state of charge based on Arrays available for operation, including both actively engaged and available but not currently engaged Arrays.

*DC-Configured Battery Systems Only.

**AC-Configured Battery Systems Only.

† Recommendation: Use the individual Modbus registers (e.g. mode, state, status) rather than this bitfield register. The bitfield is not recommended for general use, as the underlying registers provide clearer intent and easier interpretation.

1 BMS Alert Register

The register below outlines the BMS alerts at the system level. Alerts pertaining to the Arrays and their Strings can be found in the BMS Array Interface section.

Bit	Alert Name	Description
1	No Array Detected	Communication with all Arrays have been lost.
2	No Array Operational	All Arrays are not operational.
3	System Self-Disconnect Fault	The System has self-disconnected to prevent damage.
4	System Connection Lost	The System has become disconnected as a result of the last remaining Array disconnecting.
5	Array Comms Lost	Communication with one or more Arrays has been lost.

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Bit	Alert Name	Description
6	No Active Controller	Sustained communication loss with Controller.
7 to 32	Unused	

2 BMS Aggregated Array Alerts

Bit	Alert Name	Description
1 - 30	Unused	
31	Array Warning Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections. It serves as a comprehensive catch-all for any additional alerts that may arise.
32	Array Fault Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections.

BMS Extended Status Registers

Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32000	2000	AC Maximum Freq.**	Maximum AC frequency reported in the Battery System.	Hz	uint16	x100	
32001	2001	AC Minimum Freq.**	Minimum AC frequency reported in the Battery System.	Hz	uint16	x100	
32002	2002	AC Maximum Voltage L1**	Maximum line-to-neutral AC voltage level for Phase L1.	V_{RMS}	uint16	x10	
32003	2003	AC Minimum Voltage L1**	Minimum line-to-neutral AC voltage level for Phase L1.	V_{RMS}	uint16	x10	
32004	2004	AC Maximum Voltage L2**	Maximum line-to-neutral AC voltage level for Phase L2.	V_{RMS}	uint16	x10	
32005	2005	AC Minimum Voltage L2**	Minimum line-to-neutral AC voltage level for Phase L2.	V_{RMS}	uint16	x10	

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32006	2006	AC Maximum Voltage L3**	Maximum line-to-neutral AC voltage level for Phase L3.	V_{RMS}	uint16	x10	
32007	2007	AC Minimum Voltage L3**	Minimum line-to-neutral AC voltage level for Phase L3.	V_{RMS}	uint16	x10	
32008	2008	AC Maximum Voltage L1-2**	Maximum line-to-line AC voltage level for Phase L1-2.	V_{RMS}	uint16	x10	
32009	2009	AC Minimum Voltage L1-2**	Minimum line-to-line AC voltage level for Phase L1-2.	V_{RMS}	uint16	x10	
32010	2010	AC Maximum Voltage L2-3**	Maximum line-to-line AC voltage level for Phase L2-3.	V_{RMS}	uint16	x10	
32011	2011	AC Minimum Voltage L2-3**	Minimum line-to-line AC voltage level for Phase L2-3.	V_{RMS}	uint16	x10	
32012	2012	AC Maximum Voltage L3-1**	Maximum line-to-line AC voltage level for Phase L3-1.	V_{RMS}	uint16	x10	
32013	2013	AC Minimum Voltage L3-1**	Minimum line-to-line AC voltage level for Phase L3-1.	V_{RMS}	uint16	x10	
32014	2014	AC Maximum Current L1**	Maximum AC current level for Phase L1.	A_{RMS}	int16	x10	
32015	2015	AC Minimum Current L1**	Minimum AC current level for Phase L1.	A_{RMS}	int16	x10	
32016	2016	AC Maximum Current L2**	Maximum AC current level for Phase L2.	A_{RMS}	int16	x10	
32017	2017	AC Minimum Current L2**	Minimum AC current level for Phase L2.	A_{RMS}	int16	x10	
32018	2018	AC Maximum Current L3**	Maximum AC current level for Phase L3.	A_{RMS}	int16	x10	

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32019	2019	AC Minimum Current L3**	Minimum AC current level for Phase L3.	A_{RMS}	int16	x10	
32020 … 32099	2020 … 2099	Reserved					
32100	2100	Array 1 DC Voltage*	Indicates the Array voltage.	V	uint16	x10	
32101	2101	Array 1 DC Current*	Indicates the Array current.	A	int32	x10	
32103	2103	Array 1 AC Voltage L1**	Indicates the line-to-neutral AC voltage level L1.	V_{RMS}	uint16	x10	
32104	2104	Array 1 AC Voltage L2**	Indicates the line-to-neutral AC voltage level L2.	V_{RMS}	uint16	x10	
32105	2105	Array 1 AC Voltage L3**	Indicates the line-to-neutral AC voltage level L2.	V_{RMS}	uint16	x10	
32106	2106	Array 1 AC Voltage L1-2**	Indicates the line-to-line AC voltage level L1-2.	V_{RMS}	uint16	x10	
32107	2107	Array 1 AC Voltage L2-3**	Indicates the line-to-line AC voltage level L2-3.	V_{RMS}	uint16	x10	
32108	2108	Array 1 AC Voltage L3-1**	Indicates the line-to-line AC voltage level L3-1.	V_{RMS}	uint16	x10	
32109	2109	Array 1 AC Current L1**	Indicates the AC Current level for L1.	A_{RMS}	int16	x10	
32110	2110	Array 1 AC Current L2**	Indicates the AC Current level for L2.	A_{RMS}	int16	x10	

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32111	2111	Array 1 AC Current L3**	Indicates the AC Current level for L3.	A_{RMS}	int16	x10	
32112	2112	Array 1 Freq.**	Indicates the AC Frequency.	Hz	uint16	x100	
32113 … 32119	2113 … 2119	Array 1 Reserved					

*DC-Configured Battery Systems Only.

**AC-Configured Battery Systems Only.

BMS Control Registers

Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
41000	1000	Operating Command	Command to control the Battery system operation.	Enum	uint16	n/a	0: Off 1: Standby 2: On
41001	1001	Clear Alerts	Command to clear all Battery system alerts.	n/a	uint16	n/a	1: Clear All Alerts
41002	1002	Operator Heartbeat	Signal sent by the Controller to indicate it is still present.	Num	uint16	x1	Shall be sent at a frequency such that the comms lost timeout configured never expires.
41003	1003	Charge Readiness Power Command*	Charge power level that the Battery system is required to be ready for and support. Issued in advanced for the Battery System to prepare and adjust its operation. Affects efficiency as parts of the system will be brought into operation.				

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
41005	1005	Discharge Readiness Power Command*	Discharge power level that the Battery System is required to be ready for and support. Issued in advanced for the Battery System to prepare and adjust its operation. Affects efficiency as parts of the system will be brought into operation.	kW	int32	x1	$P_{Discharge} > 0$
41007	1007	DC Power Target*	Power level that the Battery System will be subject to. Allows the Battery to adjust its operation efficiently to support the current operation.	kW	int32	x1	$P_{Charge} < 0 < P_{Discharge}$
41009	1009	Active Power Setpoint**	Immediate active power level requested.	kW	int32	x1	$P > 0 \rightarrow Exporting/Discharging$ $P < 0 \rightarrow Importing/Charging$
41011	1011	Reactive Power Setpoint**	Immediate reactive power level requested.	kVAR	int32	x1	
41013	1013	Power Factor Setpoint**	System-level ratio of active power to apparent power commanded.	Num	int16	x100	$PF < 0 \rightarrow Leading$ $PF > 0 \rightarrow Lagging$
41014	1014	Operating Charge Power Limit Permitted	Maximum power level that the Battery System is allowed sink during charging.	kW	int32	x1	$P_{Charge} < 0$

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
41016	1016	Operating Discharge Power Limit Permitted	Maximum power level that the Battery System is allowed source during discharging.	kW	int32	x1	$P_{Discharge} > 0$
41018	1018	Grid Control Scope Select	Select whether system-level grid/ active/reactive mode commands are applied globally or overridden by per-Array commands.	Enum	uint16	n/a	0: System-Level (use system Grid/Active/Reactive mode commands) 1: Per-Array (ignore system mode commands; use per-Array mode commands)
41019	1019	Grid Mode Command	Commanded high-level grid interaction mode for the Battery System.	Enum	uint16	n/a	0: Invalid 1: Grid-Feeding
41020	1020	Active Power Control Mode Command	Commanded active power control mode for the Battery System.	Enum	uint16	n/a	0: Invalid 1: Fixed Active Power Control 2: Active Power Control with P-f Droop
41021	1021	Reactive Power Control Mode Command	Commanded reactive power control mode for the Battery System.	Enum	uint16	n/a	0: Invalid 1: Fixed Reactive Power (Q) Control 2: Fixed Power Factor (PF) Control

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7.2.3 BMS Array Interface

BMS Array Status Registers

Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
31000	1000	Operating State	Operating state of the Array.	Enum	int16	n/a	0: Init 1: Off 2: Standby 3: Starting Up 4: Ready (to Connect) 5: Connected 6: Transitioning to Standby 7: Turning Off 8: Service 9: Fault 10: Connecting 11: Disconnecting 12: Shutting Down
31001	1001	Operating Status	Operating status of the Array.	Enum	int16	n/a	0: Invalid 1: Normal 2: Warning 3: Fault 4: Critical Fault
31002	1002	Operating Mode	Operating mode of the Battery System.	Enum	int16	n/a	0: Invalid 1: Non-Operational 2: Operational 3: Service 4: Initial Charge

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
31003	1003	Active State of Charge	Active state of charge of the Array.	%	uint16	x10	0 - 100% Array state of charge based on the Strings actively engaged in operation (excludes available but inactive Strings).
31004	1004	String Maximum State of Charge	Indicates the highest State of Charge among the Array Strings.	%	uint16	x10	0 - 100%
31005	1005	String Minimum State of Charge	Indicates the lowest State of Charge among the Array Strings.	%	uint16	x10	0 - 100%
31006	1006	DC Power*	Total DC power reported by the Strings.	kW	int32	x1	$P > 0 \rightarrow Discharging$ $P < 0 \rightarrow Charging$
31008	1008	Active Power**	Indicates the total Array AC active power reported by the PCS devices.	kW	int32	x1	$P > 0 \rightarrow Exporting/Discharging$ $P < 0 \rightarrow Importing/Charging$
31010	1010	Reactive Power**	Indicates the total Array AC reactive power reported by the PCS devices.	kVAR	int32	x1	$Q > 0 \rightarrow Inductive\ Power$ $Q < 0 \rightarrow Capacitive\ Power$
31012	1012	Power Factor**	Array-level ratio of active power to apparent power.	Num	int16	x100	$PF < 0 \rightarrow Leading$ $PF > 0 \rightarrow Lagging$
31013	1013	Discharge Energy	Indicates the energy level of the Array available for discharge.	kWh	uint32	x1	
31015	1015	Charge Energy	Indicates the energy level required to reach Top-of-Charge in the Array.	kWh	uint32	x1	

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
31017	1017	Charge Power Limit	Indicates the charge power immediately available from the Array.	kW	int32	x1	$P_{Charge} < 0$
31019	1019	Discharge Power Limit	Indicates the discharge power immediately available from the Array.	kW	int32	x1	$P_{Discharge} > 0$
31021	1021	Potential Discharge Energy	Indicates the energy level of the Array that can be available within a short period of time.	kWh	int32	x1	
31023	1023	Potential Charge Energy	Indicates the energy level required to reach Top-of-Charge that can be available from the Array within a short period of time.	kWh	int32	x1	
31025	1025	Potential Charge Power Limit	Indicates the charge power that can be available from the Array within a short period of time.	kW	int32	x1	$P_{Charge} < 0$
31027	1027	Potential Discharge Power Limit	Indicates the discharge power that can be available from the Array within a short period of time.	kW	int32	x1	$P_{Discharge} > 0$
31029	1029	Average Electrolyte Temp.	Average Electrolyte Temperature in the Array.	°C	int16	x10	
31030	1030	Minimum Electrolyte Temp.	Minimum Electrolyte Temperature in the Array.	°C	int16	x10	

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
31031	1031	Maximum Electrolyte Temp.	Maximum Electrolyte Temperature in the Array.	°C	int16	x10	
31032	1032	String Connected Count	Number of Strings actively supporting the operation.	Count	uint16	x1	
31033	1033	String Available Count	Number of Strings available to support the operation including those actively doing so.	Count	uint16	x1	
31034	1034	Control Mode	The control mode indicates whether the Array is being controlled from the Parent Controller (System Controller) or in Service under control of Service Personnel from the Service interface.	Enum	uint16	n/a	0: Normal 1: User-Controlled
31035	1035	Heartbeat	BMS Array Heartbeat.	Num	uint16	x1	Incremented every 1s and wraps around on overflow.
31036	1036	Array-Specific Alerts	Provides a summary of alert states at the Array level.	Bitfield	uint32	n/a	1: active alert 0: inactive alert
31038	1038	Array Aggregated String Alerts	Provides a summary of alert states at the Array level, resulting from the aggregation of String alerts.	Bitfield	uint32	n/a	1: active alert 0: inactive alert
31040	1040	String Alert Change Summary	Provide a summary of which strings changes in their alert states.	Bitfield	uint32	n/a	1: at least one String alert has changed its state for that particular String indicated by the bit position. 0: no alert is active

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Reg (dec)	Addr (dec)	Register	Description	Units	Data Type	Scaling Factor	Notes
31042	1042	String 1 Alerts	Provides a summary of alert states for String #1.	Bitfield	uint32	n/a	1: active alert 0: inactive alert
		…					
31080	1080	String 20 Alerts	Provides a summary of alert states for String #20.	Bitfield	uint32	n/a	1: active alert 0: inactive alert
31100 - 31115	1100 - 1115	PCS Alert 1*** - PCS Alert 16***	16 alert data registers reported by the PCS. The structure of these registers varies depending on the PCS model.	Bitfield	uint16	x1	Each PCS Alert register is a 16-bit bitfield. Bit 0 is the least-significant bit and bit 15 is the most-significant bit. 1: active alert 0: inactive alert
31200	1200	Available State of Charge	Available state of charge of the Array.	%	uint16	x10	0 - 100% Array state of charge based on Strings available for operation, including both actively engaged and available but not currently engaged Strings.

1 Array-Specific Alert Register

Bit	Alert Name	Description
1	No String Detected	Communication with all Strings have been lost.
2	No String Operational	All Array Strings are not operational.
3	Array Self-Disconnect Fault	The Array has self-disconnected to prevent damage.
4	Array Connection Lost	The Array has become disconnected as a result of the last remaining String disconnecting.
5	String Comms Lost	Communication with one or more Strings has been lost.
6	Active Power Level Request Above Charge Limit	The requested active power level for charging exceeds the maximum charge capability of the Array.
7	Active Power Level Request Above Discharge Limit	The requested active power level for discharging exceeds the maximum discharge capability of the Array.

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Bit	Alert Name	Description
8	Support Power Level Request Above Charge Limit	The requested support power level for charging exceeds the available charge capability of the Array.
9	Support Power Level Request Above Discharge Limit	The requested support power level for discharging exceeds the available discharge capability of the Array.
10	No Active Controller	Sustained communication loss with Controller.
11 to 32	Unused	

2 Array Aggregated String Alerts

Bit	Alert Name	Description
1	High Temperature Warning Active	A String has reported temperatures above normal operating levels.
2	High Temperature Fault Active	A String has reported temperatures exceeding the maximum threshold.
3	Low Temperature Warning Active	A String has reported temperatures below normal operating levels.
4	Low Temperature Fault Active	A String has reported temperatures below the minimum threshold.
5	Subsystem Warning Active	A String has reported a subsystem warning.
6	Subsystem Fault Active	A String has reported a subsystem fault.
7	Power Distribution Fault Active	A String has reported a power-related fault.
8	Electrolyte SoC Out Of Range	A String has reported electrolyte state of charge out of range.
9	Safety Mechanism Active	A String has reported a safety mechanism active.
10	String Over Current Fault	A String has reported an overcurrent fault condition.
11	String Current Limit Exceeded	A String has reported a current limit exceeded condition
12 - 30	Unused	
31	String Warning Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections. It serves as a comprehensive catch-all for any additional alerts that may arise.
32	String Fault Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections.

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3 String Alert Registers

Bit	Alert Name	Description
1	Module Operation Fault	One or More String Modules are operating outside the expected state.
2	Module Communication Loss	One or more modules in the string have stopped communicating.
3	Electrolyte Temperature High Fault	Electrolyte temperature maximum level exceeded.
4	Electrolyte Temperature Low Fault	Electrolyte temperature minimum level exceeded.
5	Electrolyte SoC High	Electrolyte SoC is above maximum threshold.
6	Electrolyte SoC Low	Electrolyte SoC is below minimum threshold.
7	Pump Subsystem Failure	A Module has reported a failure in the electrolyte pump subsystem.
8	Fan Subsystem Failure	A Module has reported a failure in the fan subsystem.
9	Stack Subsystem Failure	A Module has reported a failure in the stack subsystem.
10	Module Safety Mechanism Active	A Module has reported an active safety mechanism.
11	Module Subsystem Circuit Breaker Opened Fault	A Module has reported a subsystem circuit breaker opening fault.
12	Module Power Failure	A Module has reported a failure in the 3-phase power supply.
13	High SoC / High Temperature Operating Time Fault	Total accumulated operating time at High SoC / High Temperature is above max allowed.
14	Charge Current Limit Exceeded	Charging current level exceeded maximum allowed.
15	Charge Over Current Fault	The charge current is too high.
16	Discharge Current Limit Exceeded	Discharging current level exceeded maximum allowed.
17	Discharge Over Current Fault	The discharge current is too high.
18	String Over Voltage	String voltage is above the maximum threshold.
19	String Repeated Startup Failure	One or more modules in the String failed to turn on repeatedly.
20	Positive String Contactor Failed	Positive string contactor operation has failed.
21	Negative String Contactor Failed	Negative string contactor operation has failed.
22	Self-Disconnect Fault	The <component> has self-disconnected to prevent damage.

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Bit	Alert Name	Description
23	Module Internal Fault	One or More Modules' have reported an internal failure.
24	E-stop Active	The E-stop has been activated.
25	Manual Disconnect Opened	Manual disconnect detected open.
26	Module Circuit Breaker Opened Fault	A Module circuit breaker has opened.
27 - 30	Unused	
31	String Warning Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections. It serves as a comprehensive catch-all for any additional alerts that may arise.
32	String Fault Active	Catch-all register alerts that have not been grouped or aggregated in the previous sections.

*DC-Configured Battery Systems Only.

**AC-Configured Battery Systems Only.

4 PCS Alert Registers

The PCS alert registers are PCS-specific. Their structure and meaning can vary depending on the PCS type and model.

***Gamesa PCS Proteus 3150E

The tables below provide the Gamesa PCS Proteus 3150E mapping where known.

Register	Bit	Name	Description	Bit	Name	Description
PCS Alert 1 - Active Alarms 1 for the PCS	0	GRND FLT	Ground fault alarm.	8	OVER FREC	Over frequency alarm.
	1	DC OV	DC over voltage alarm.	9	UNDER FREC	Under frequency alarm.
	2	AC DISC	AC disconnected alarm.	10	OVER VOLT	AC over voltage alarm.
	3	DC DISC	DC disconnected alarm.	11	UNDER VOLT	AC under voltage alarm.
	4	GRID DISC	Grid disconnected alarm.	12	--	--
	5	--	--	13	UNDER TEMP	Under temperature alarm.
	6	EMERG	Emergency relay alarm.	14	--	--
	7	OVER TEMP	Over temperature alarm.	15	HW FLT	Hardware failure alarm.

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Register	Bit	Name	Description	Bit	Name	Description
PCS Alert 2 - Active Alarms 2 for the PCS	0	COMM	Communication alarm.	8	--	--
	1	INT	PCS internal alarm.	9	--	--
	2	MAINT	Maintenance alarm.	10	--	--
	3	BATT	Battery alarm.	11	--	--
	4	PW LIM	Power limitation alarm.	12	--	--
	5	--	--	13	--	--
	6	--	--	14	--	--
	7	--	--	15	--	--
PCS Alert 3 - Active Warning 1 for the PCS	0	GRND FLT	Ground fault warning.	8	OVER FREC	Over frequency warning.
	1	DC OV	DC over voltage warning.	9	UNDER FREC	Under frequency warning.
	2	AC DISC	AC disconnected warning.	10	OVER VOLT	AC over voltage warning.
	3	DC DISC	DC disconnected warning.	11	UNDER VOLT	AC under voltage warning.
	4	GRID DISC	Grid disconnected warning.	12	--	--
	5	--	--	13	UNDER TEMP	Under temperature warning.
	6	EMERG	Emergency relay warning.	14	--	--
	7	OVER TEMP	Over temperature warning.	15	HW FLT	Hardware failure warning.
PCS Alert 4 - Active Alarms 2 for the PCS	0	COMM	Communication warning.	8	--	--
	1	INT	PCS internal warning.	9	--	--
	2	MAINT	Maintenance warning.	10	--	--
	3	BATT	Battery warning.	11	--	--
	4	PW LIM	Power limitation warning.	12	--	--
	5	--	--	13	--	--

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Register	Bit	Name	Description	Bit	Name	Description
	6	--	--	14	--	--
	7	--	--	15	--	--
PCS Alert 5 - Active Alarms 1 for the power section 1	0	GRND FLT	Ground fault alarm.	8	--	--
	1	DC OV	DC over voltage alarm.	9	--	--
	2	--	--	10	--	--
	3	DC DISC	DC disconnected alarm.	11	--	--
	4	--	--	12	--	--
	5	--	--	13	--	--
	6	EMERG	Emergency relay alarm.	14	--	--
	7	--	--	15	HW FLT	Hardware failure alarm.
PCS Alert 6 - Active Alarms 2 for the power section 1	0	COMM	Communication alarm.	8	DISABLED	Power module disabled alarm.
	1	--		9	--	--
	2	--		10	--	--
	3	BATT	Battery alarm.	11	--	--
	4	PW LIM	Power limitation alarm.	12	--	--
	5	RSVS	Reverse polarity alarm.	13	--	--
	6	DC OC	DC over current alarm.	14	--	--
	7	DSCR	DC current discrepancy alarm.	15	--	--
PCS Alert 7 - Active Warning 1 for the power section 1	0	GRND FLT	Ground fault warning.	8	--	--
	1	DC OV	DC over voltage warning.	9	--	--
	2	--	--	10	--	--
	3	DC DISC	DC disconnected warning.	11	--	--
	4	--	--	12	--	--
	5	--	--	13	--	--

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Register	Bit	Name	Description	Bit	Name	Description
	6	EMERG	Emergency relay warning.	14	--	--
	7	--	--	15	HW FLT	Hardware failure warning.
PCS Alert 8 - Active Warning 2 for the power section 1	0	COMM	Communication warning.	8	DISABLED	Power module disabled warning.
	1	--		9	--	--
	2	--		10	--	--
	3	BATT	Battery warning.	11	--	--
	4	PW LIM	Power limitation warning.	12	--	--
	5	RSVS	Reverse polarity warning.	13	--	--
	6	DC OC	DC over current warning.	14	--	--
	7	DSCR	DC current discrepancy warning.	15	--	--
PCS Alert 9, 10	Similar to PCS alert 5, 6 but for the power section 2.					
PCS Alert 11, 12	Similar to PCS alert 7, 8 but for the power section 2.					

BMS Array Extended Status Registers

Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32000	2000	DC Voltage	Indicates the Array voltage.	V	uint16	x10	
32001	2001	DC Current	Indicates the Array current.	A	int32	x10	
32003	2003	AC Voltage L1**	Indicates the line-to-neutral AC voltage level L1.	V_{RMS}	uint16	x10	
32004	2004	AC Voltage L2**	Indicates the line-to-neutral AC voltage level L2.	V_{RMS}	uint16	x10	

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
32005	2005	AC Voltage L3**	Indicates the line-to-neutral AC voltage level L2.	V_{RMS}	uint16	x10	
32006	2006	AC Voltage L1-2**	Indicates the line-to-line AC voltage level L1-2.	V_{RMS}	uint16	x10	
32007	2007	AC Voltage L2-3**	Indicates the line-to-line AC voltage level L2-3.	V_{RMS}	uint16	x10	
32008	2008	AC Voltage L3-1**	Indicates the line-to-line AC voltage level L3-1.	V_{RMS}	uint16	x10	
32009	2009	AC Current L1**	Indicates the AC Current level for L1.	A_{RMS}	int16	x10	
32010	2010	AC Current L2**	Indicates the AC Current level for L2.	A_{RMS}	int16	x10	
32011	2011	AC Current L3**	Indicates the AC Current level for L3.	A_{RMS}	int16	x10	
32012	2012	Freq.**	Indicates the AC Frequency.	Hz	uint16	x100	

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**AC-Configured Battery Systems Only.

BMS Array Service Registers

Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
33000	3000	PCS Operating Mode Request**	PCS operational required for initial charge phase.	Enum	int16	n/a	0: Not required 1: Power Mode 2: Voltage Mode
33001	3001	PCS Voltage Mode Minimum Voltage**	Voltage mode minimum voltage level requested to the PCS.	V	uint16	x1	

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
33002	3002	PCS Voltage Mode Maximum Voltage**	Voltage mode maximum voltage level requested to the PCS.	V	uint16	x1	
33003	3003	PCS Power Mode Active Power Request**	Power mode active power setpoint required from the PCS.	kW	int32	x1	$P > 0 \rightarrow Exporting/Discharging$ $P < 0 \rightarrow Importing/Charging$

*DC-Configured Battery Systems Only.

**AC-Configured Battery Systems Only.

BMS Array Control Registers

Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
41000	1000	Array Upper SOC Limit	Max. SOC value allowed for the Array Strings.	%	uint16	x10	0 - 100%
41001	1001	Array Lower SOC Limit	Min. SOC value allowed for the Array Strings.	%	uint16	x10	0 - 100%
41002	1002	Operating Charge Power Limit Permitted	Maximum power level that the Array is allowed sink during charging.	kW	int32	x1	$P_{Charge} < 0$
41004	1004	Operating Discharge Power Limit Permitted	Maximum power level that the Array is allowed source during discharging.	kW	int32	x1	$P_{Discharge} > 0$
41006	1006	Operator Heartbeat	Signal sent by the Controller to indicate it is still present.	Num	uint16	x1	Shall be sent at a frequency such that the comms lost timeout configured never expires.
41007	1007	Grid Mode Command	Commanded high-level grid interaction mode for the Array.	Enum	uint16	n/a	0: Invalid 1: Grid-Feeding

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Reg (dec)	Addr (dec)	Name	Description	Units	Data Type	Scaling Factor	Notes
41008	1008	Active Power Control Mode Command	Commanded active power control mode for the Array.	Enum	uint16	n/a	0: Invalid 1: Fixed Active Power Control 2: Active Power Control with P-f Droop
41009	1009	Reactive Power Control Mode Command	Commanded reactive power control mode for the Array.	Enum	uint16	n/a	0: Invalid 1: Fixed Reactive Power (Q) Control 2: Fixed Power Factor (PF) Control

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